

Datasets Used

1. Healthcare & Medical – Sample Datasets

a. Breast Cancer (Synthetic Sample)

radius_mean	texture_mean	smoothness	concavity	symmetry	label
14.2	19.4	0.10	0.12	0.18	malignant
12.1	16.3	0.09	0.05	0.17	benign
15.4	22.7	0.11	0.16	0.20	malignant
11.8	14.8	0.08	0.03	0.15	benign

b. PIMA Diabetes (Synthetic Sample)

pregnancies	glucose	insulin	bmi	age	outcome
2	130	0	32.1	45	1
5	95	85	27.8	31	0
1	180	120	36.4	50	1
3	110	0	29.0	28	0

c. MIMIC-III (Mini Slice Sample)

patient_id	heart_rate	spo2	systolic_bp	diagnosis	stay_length_days
101	98	92	130	sepsis	5
102	82	97	118	cardiac arrest	8
103	120	88	145	pneumonia	4

d. Heart Disease Dataset (Synthetic)

age	cholesterol	resting_ecg	max_hr	exercise_angina	target
56	240	normal	150	no	0
63	310	hypertrophy	138	yes	1

	age	cholesterol	resting_ecg	max_hr	exercise_angina	target
	44	200	normal	170	no	0

2. Finance & Banking – Sample Datasets

a. Credit Card Fraud (Synthetic)

	amount	time	V1	V2	V3	fraud
	120.50	3000	-1.23	0.78	-0.12	0
	560.10	5200	2.10	-3.50	1.22	1
	80.40	1800	-0.50	1.12	-0.45	0

b. German Credit (Synthetic)

	age	job	credit_amount	housing	duration_months	label
	35	skilled	1200	own	12	good
	48	unskilled	2500	rent	24	bad
	29	skilled	800	free	10	good

c. Stock Prices (Synthetic)

	date	open	high	low	close	volume
	2025-01-01	100	105	98	103	1,200,000
	2025-01-02	103	108	102	107	1,500,000
	2025-01-03	107	109	105	106	900,000

d. LendingClub Loans (Synthetic)

loan_amount interest_rate employment_length income default

10,000	12.5%	5 years	65,000	0
25,000	20.1%	1 year	40,000	1
15,000	16.0%	3 years	52,000	0

3. Retail & E-Commerce – Sample Datasets

a. Online Retail Transactions (Synthetic)

invoice_no product quantity price timestamp

1001	T-Shirt	2	499	2025-01-03 10:21
1002	Mug	-1	199	2025-01-03 11:15
1003	Shoes	1	1999	2025-01-03 14:30

b. Amazon Reviews (Synthetic)

review_text rating product_category

"Great headphones, amazing bass!"	5	electronics
"Terrible quality, broke in 2 days."	1	electronics
"Good for the price."	4	home

c. Groceries Basket (Synthetic)

items

milk, bread, butter

apples, bananas

bread, eggs, milk

d. Instacart (Synthetic)

order_id product_id user_id add_to_cart_order reordered

1	101	501	1	0
1	205	501	2	1
2	110	600	1	0

4. NLP – Sample Text Datasets

a. IMDB Sentiment (Synthetic)

review sentiment

“A masterpiece, loved every scene.” positive

“Worst film of the year.” negative

b. SMS Spam (Synthetic)

message label

“Win ₹10,000 NOW! Click link” spam

“Meeting at 5 PM?” ham

c. News Categories (Synthetic)

headline category

“Budget raises taxes on imports” politics

“Team X wins championship” sports

d. SQuAD (Mini Synthetic)

context question answer

“The sun is a star located at the center of the solar system.” What is the sun? a star

5. Computer Vision – Sample Image Dataset (Tabular Representation)

a. MNIST-like Pixels (Synthetic)

pixel1	pixel2	pixel3	...	label
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0	255	128	...	7
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200	60	0	...	3
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b. CIFAR-10 (Metadata Sample)

image_id	class	resolution
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img001	airplane	32×32×3
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img002	cat	32×32×3
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c. COCO (Detection Sample)

image	object	x_min	y_min	x_max	y_max
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img01	person	34	50	120	200
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img01	dog	140	60	300	180
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6. Recommendation Systems – Sample Dataset

a. MovieLens (Synthetic)

user_id	movie_id	rating	timestamp
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1	101	5	162000
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2	210	3	162500
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1	210	4	163000
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b. Goodbooks-10K (Synthetic)

user_id	book_id	rating
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10	550	5
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user_id	book_id	rating
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8	120	3
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7. General ML – Sample Datasets

a. Iris (Synthetic)

sepal_len	sepal_wid	petal_len	petal_wid	species
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5.1	3.5	1.4	0.2	setosa
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6.7	3.0	5.2	2.3	virginica
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b. Housing Price (Synthetic)

rooms	distance	crime_rate	price
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5	2.3	0.04	320,000
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8	5.8	0.15	190,000
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c. Titanic (Synthetic)

sex	age	class	fare	survived
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male	22	3	7.25	0
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female	38	1	71.28	1
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d. Adult Census Income (Synthetic)

age	education	hours_per_week	income
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35	Bachelors	40	>50K
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50	HS-grad	30	≤50K
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8. Cybersecurity – Sample Datasets

a. NSL-KDD (Synthetic)

	duration	protocol	service	src_bytes	dst_bytes	label
0		tcp	http	181	5450	normal
2		udp	domain 0		32	neptune

b. CICIDS2017 (Synthetic)

flow_duration	total_fwd_pkts	total_bwd_pkts	src_ip	dst_ip	attack
200000	15	3	10.0.0.1	10.0.0.9	benign
600000	200	5	10.0.0.5	10.0.0.20	DoS

9. Social Media Analytics – Sample Datasets

a. Twitter Sentiment140 (Synthetic)

tweet	sentiment
“Feeling amazing today 😊”	positive
“Traffic jam ruined my day...”	negative

b. Reddit Comments (Synthetic)

comment	subreddit	toxicity_score
“This explanation really helped me!”	r/learnpython	0.02
“You are completely wrong idiot”	r/gaming	0.78

10. Environmental & Geospatial – Sample Datasets

a. Air Quality UCI (Synthetic)

CO	NO2	temp	humidity	pressure	time
1.5	40	22.5	48	1012	10:00

CO NO2 temp humidity pressure time

0.8 28 25.0 52 1010 11:00

b. NASA Earth Observations (Synthetic)

lat lon vegetation_index land_temp

34.5 -118.3 0.42 295

12.1 77.6 0.56 301

c. Earthquake Dataset (Synthetic)

magnitude depth latitude longitude date

4.8 10 38.1 142.9 2025-01-01

6.2 25 -3.1 135.0 2025-02-10